



Hypothesis Testing for Poisson and Binomial Parameters · Mark Scheme

Further Statistics 1 Chapter 4: Hypothesis Testing for Poisson and Binomial Parameters. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Tests for a Poisson mean

- 1 $H_0: \lambda = 5$ B1; $H_1: \lambda > 5$, where λ is the mean number of emails per hour B1.
- 2 $P(X \geq 8) = 1 - P(X \leq 7)$ M1; $= 0.0511$ A1; $0.0511 > 0.05$, so do not reject H_0 : insufficient evidence that $\lambda > 4$ A1.
- 3 $P(X \leq 1) = e^{-6}(1 + 6)$ M1; $= 0.0174$ A1; $0.0174 < 0.05$, so reject H_0 : evidence that $\lambda < 6$ A1.
- 4 $H_0: \lambda = 1.5$, $H_1: \lambda < 1.5$ (per month) B1; in 3 months $X \sim \text{Po}(4.5)$ M1; $P(X \leq 1) = 0.0611$ A1; $0.0611 > 0.05$, not significant: insufficient evidence that the rate of accidents has decreased A1.
- 5(a) Complaints occur singly and independently B1; at a constant average rate B1.
- 5(b) $H_0: \lambda = 3$, $H_1: \lambda < 3$ B1; under H_0 , $X \sim \text{Po}(12)$ M1; $P(X \leq 6) = 0.0458$ A1; $0.0458 < 0.05$, reject H_0 : there is evidence that the rate of complaints has decreased A1.
- 5(c) $P(X \leq 6) = 0.0458 < 0.05$ and $P(X \leq 7) = 0.0895 > 0.05$ M1; largest number is 6 A1.

Section B – Critical regions

- 6 $P(X \geq 10) = 0.0193$ and $P(X \geq 9) = 0.0730$ M1; critical region $X \geq 10$ A1.
- 7 $P(X \geq 10) = 1 - 0.9682 = 0.0318$, $P(X \geq 9) = 0.0681$ M1; critical region $X \geq 10$ A1; actual significance level 3.18% A1.
- 8 $P(X \leq 2) = 0.0355$, $P(X \leq 3) = 0.1071$ M1; critical region $X \leq 2$ A1; actual significance level 3.55% A1.
- 9 Under H_0 , $X \sim \text{Po}(6)$ B1; $P(X \geq 11) = 1 - 0.9574 = 0.0426$, $P(X \geq 10) = 0.0839$ M1; critical region $X \geq 11$ A1; actual significance level 4.26% A1.
- 10(a) p = proportion of faulty phone cases B1; $H_0: p = 0.02$, $H_1: p > 0.02$ B1.
- 10(b) n large and p small, so $X \approx \text{Po}(150 \times 0.02) = \text{Po}(3)$ B1; $P(X \geq 7) = 1 - 0.9665 = 0.0335$ M1; $P(X \geq 6) = 0.0839 > 0.05$ A1; critical region $X \geq 7$ A1.
- 10(c) 3.35% B1.
- 10(d) 8 lies in the critical region, so reject H_0 M1; there is evidence that the proportion of faulty phone cases has increased A1.

Section C – Two-tailed tests and p-values

- 11 The probability, assuming H_0 is true, of a result at least as extreme as the one observed B1; reject H_0 if the p-value is less than the significance level (half of it, for each tail of a two-tailed test) B1.
- 12 $P(X \geq 9) = 1 - 0.9786 = 0.0214$ M1 A1; $0.0214 < 0.025$, reject H_0 : evidence that $\lambda \neq 4$ A1.
- 13 $P(X \leq 2) = 0.0913$ M1 A1; $0.0913 > 0.05$, do not reject H_0 : insufficient evidence that $p \neq 0.25$ A1.
- 14 Lower tail: $P(X \leq 2) = 0.0138$, $P(X \leq 3) = 0.0424$, so $X \leq 2$ M1 A1; upper tail: $P(X \geq 15) = 0.0173$, $P(X \geq 14) = 0.0342$, so $X \geq 15$ A1; actual significance level $0.0138 + 0.0173 = 3.11\%$ A1.
- 15(a) $H_0: p = 0.35$, $H_1: p \neq 0.35$ B1.
- 15(b) $X \sim \text{B}(30, 0.35)$; $P(X \leq 5) = 0.0233$, $P(X \leq 6) = 0.0586$ M1; lower tail $X \leq 5$ A1; $P(X \geq 17) = 0.0124$, $P(X \geq 16) = 0.0301$ M1; upper tail $X \geq 17$ A1.
- 15(c) $0.0233 + 0.0124 = 3.56\%$ B1.



15(d) 16 is not in the critical region, so do not reject H_0 *M1*; insufficient evidence that the probability of landing on red is not 0.35 *A1*.

Roar-level challenge

16 (a) Under H_0 , faults in t hours $\sim \text{Po}(0.5t)$, $P(X = 0) = e^{-0.5t}$ *M1*; $e^{-0.5t} \leq 0.05 \Rightarrow t \geq 2\ln 20 = 5.99$ *M1*; $t = 6$ *A1*, actual significance level $e^{-3} = 4.98\%$ *A1*. (b) $\text{Po}(6)$: $P(X \leq 1) = 7e^{-6} = 1.74\%$ *B1*.